Interval counting verification

Genesis/Python read-end overlap counts and bedtools coverage

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Abstract

Across 160 evaluated library/method/repetition records, independent read-end counts have zero mismatching regions. The records contain 892,528 region evaluations, including repeated regions across methods and technical runs. This verifies arithmetic under the declared overlap convention, not biological accuracy or relative speed. We would retain the current counter and its independent checks.

Setup

Comparisons. The benchmark counts retained primary read-end spans in zero-based half-open peak intervals using Python and bedtools 2.31.1 coverage. Paired ends count independently. Fragment-level FRiP is a different quantity and is not used as a counting-equivalence target.

Scoring. A mismatch is a region whose integer overlap counts differ. Any shared base counts as overlap; no fractional NH weighting is introduced. Denominators count evaluated region instances, not distinct regulatory elements or biological replicates.

Table 1. Read-end counting comparisons; repeated region evaluations are retained.

Cohort	Records	Region evaluations	Mismatches
dap-se-1m	24	48,888	0
dap-se-5m	8	17,876	0
dap-pe-1m	48	291,804	0
dap-pe-5m	16	130,152	0
atac-arabidopsis-1m	48	192,960	0
atac-arabidopsis-5m	16	210,848	0

1 The evaluated integer overlap counts agree exactly

The 160 evaluated records report 0 mismatches over 892,528 region instances. This denominator includes the 1-million and 5-million caps, multiple method arms, and repeated execution; none increases the number of biological samples.

Table 2. Measurement conventions distinguish otherwise similar interval files.

Quantity	Unit / denominator	Equivalent?
Read-end overlap count	Integer aligned ends per region	Tested
Fragment FRiP	Fragments overlapping peaks / retained fragments	Different target
Peak RPM	Read-end count / retained ends, scaled by 1 million	Normalization separate
Mean coverage RPKM	Length-weighted coverage, including zero bases	Different target

2 The evidence does not establish a counting speed advantage

The independent bedtools evaluation runs after processing, whereas the 64 method-case processing timings exclude common evaluation. There are 0 matched standalone Python-versus-bedtools timing series in this report, so relative speed and memory are nd.

Table 3. nd means not determined in a matched counting-only comparison.

Measurement	Python	bedtools
Matching read-end arithmetic	PASS	PASS
Matched standalone runtime	nd	nd
Matched standalone peak memory	nd	nd

Missing values. A missing or unevaluated count is not zero. Any unavailable records remain in the exported counting table with their status and reason; the denominator above includes only records explicitly marked OK.

Publication scope. These checks cover the benchmark's independent integer read-end overlap calculation. They do not certify all production quantification modes, NH fractional weighting, spliced-block conventions, or coverage integration by transitivity.

Open issues

- Repeated evaluation of a fixed peak set is not independent biological validation.
- Span-based overlap should not be generalized to spliced exon-block counting.
- No matched standalone runtime advantage is established.

Recommendation

We would retain exact arithmetic tests and avoid choosing a counter by unmatched timings.

1. Define a counting-only test with identical interval ordering, BAM cache conditions and units.
2. Measure time and process memory separately from conversion and file staging.

Appendix: Methods

We summarize recorded results without rerunning biological processing. The main Arabidopsis matrix uses seeded nested raw-template subsets (seed 60141), three fresh technical repetitions at the 1-million cap and one at the 5-million cap. Actual counts replace requested caps for short libraries. DAP retains primary mapped reads including MAPQ 0; ATAC selects proper primary pairs with both mates MAPQ at least 30, excludes MAPQ 255, organelles and marked duplicates. Biological identity is preserved. Source files are SHA-256 checked against the accompanying catalog; evidence snapshot revision ed1c50b8e4b5 includes the October 13 to 16 records. Completed local reference assessments are separately identified in that catalog.